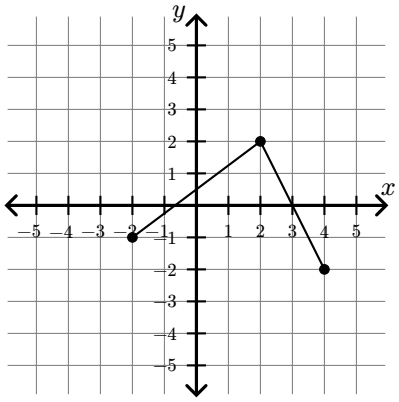
Section 6.6 — Function Transformations

Math 103: College Algebra

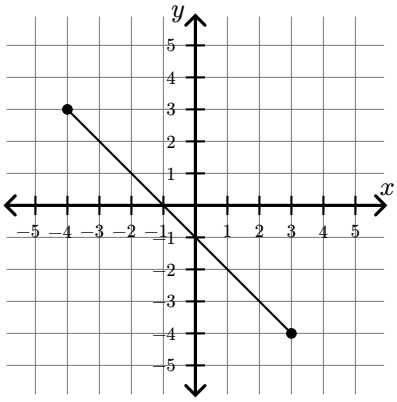
Single-Step Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points.

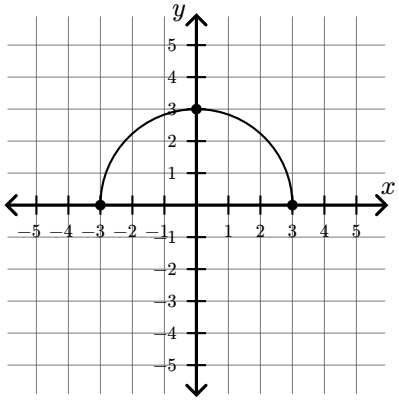
1. $g(x) = 2f(x)$



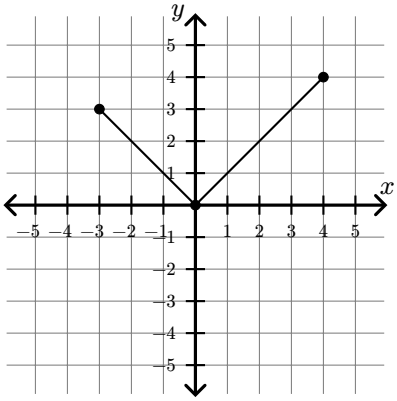
2. $g(x) = f(-x)$



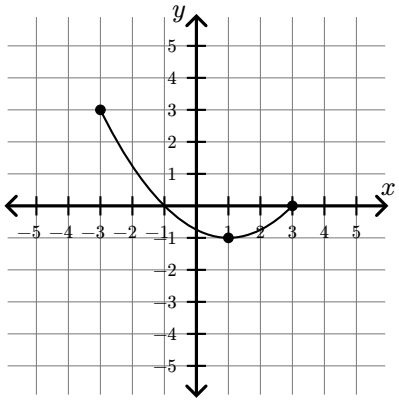
3. $g(x) = f(x) - 3$



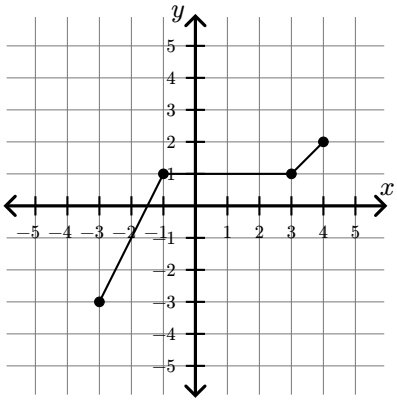
4. $g(x) = -f(x)$



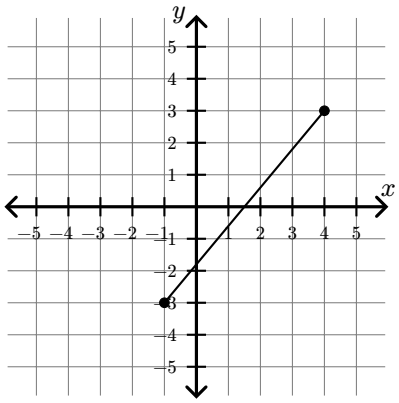
5. $g(x) = f(x) + 2$



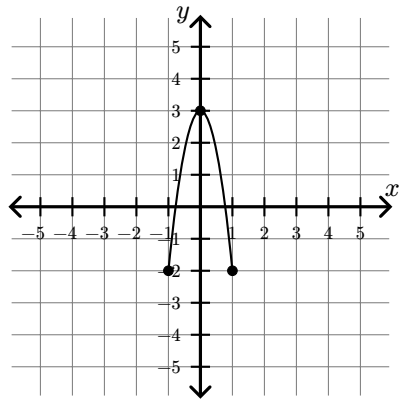
6. $g(x) = f(x + 1)$



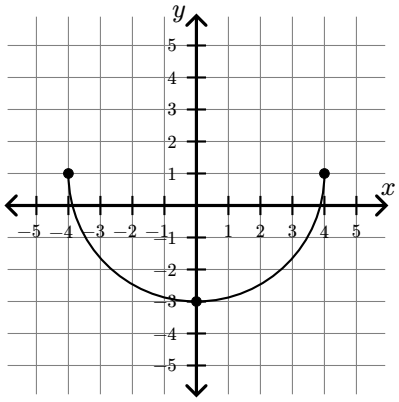
7. $g(x) = \frac{2}{3}f(x)$



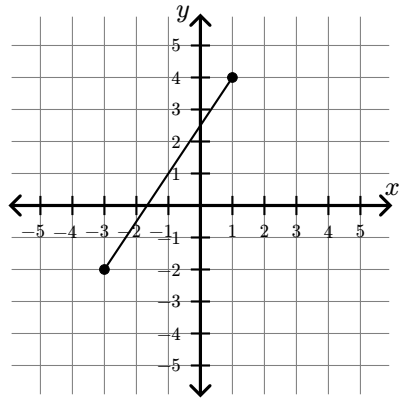
8. $g(x) = f(\frac{1}{3}x)$



9. $g(x) = f(2x)$



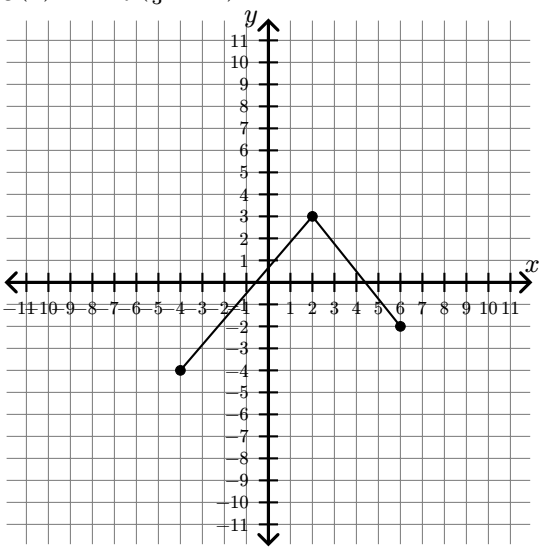
10. $g(x) = f(x-3)$



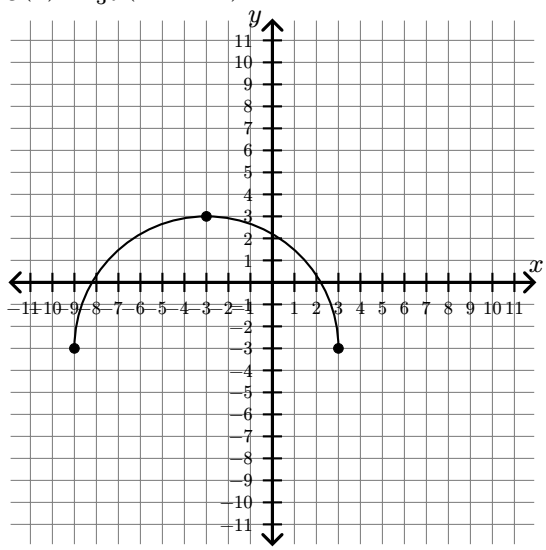
Multiple Transformations

In each figure below, the given graph is the graph of $y = f(x)$. Graph the function in the question on the same axes. You may find it helpful to use the marked points as reference points. *You will want to write out your work on a separate sheet of paper.*

11. $g(x) = -2f(\frac{2}{3}x + 2) + 3$



12. $g(x) = \frac{2}{3}f(-2x - 5) - 1$



Domain and Range

13. If $g(x) = 3f(-2x-3)+4$ and $\operatorname{Dom} f = (-5,\infty)$, what is $\operatorname{Dom} g$?

14. If $h(x) = \frac{1}{3}g(x+1)-7$ and $\operatorname{Rng} g = (-\infty,3]\cup[5,\infty)$, what is $\operatorname{Rng} h$?

15. If $k(x) = 2f\left(7-\frac{2}{5}x\right)-6$, $\operatorname{Dom} f = (0,1)$, and $\operatorname{Rng} f = (-\infty,\infty)$, what are $\operatorname{Dom} k$ and $\operatorname{Rng} k$?

Symmetry

Classify each function as even, odd, or neither.

16. $f(x) = x^2$
17. $f(x) = (x-7)^2$
18. $f(x) = x^5$
19. $f(x) = 2x$
20. $f(x) = x^6 - 3x^2 + 9$
21. $f(x) = x^3 + x^2$
22. $f(x) = x^3 + 6$
23. $f(x) = x^9 - x^3 + 8x$
24. $f(x) = \sqrt{x}$
25. $f(x) = \sqrt[5]{x}$
26. $f(x) = 3|2x| - 5$
27. $f(x) = |x-5|$
28. Based on your answers to #16 – 23, how can we decide if a polynomial is an even function, an odd function, or neither? (Try to answer this question on your own before looking at the answer key.)

Using the Library

For each function below:

a. Identify the base function (call it $f(x)$) and the family of functions it belongs to.

b. Rewrite the function in terms of $f(x)$. (We usually won't do this explicitly in practice.)

c. Sketch a graph of the function, including any asymptotes and clearly marking any special points, using a separate sheet of graph paper or the other side of this page.

d. Give the domain and range of the function.

29. $g(x) = -3x + 5$
30. $g(x) = \dfrac{5}{x^2 - 6x + 9}$
31. $g(x) = \dfrac{3}{x-2}$
32. $g(x) = |x-5| + 2$
33. $g(x) = 4$
34. $g(x) = 2(x-1)^3 + 1$
35. $g(x) = \log_{1/3}(2x+3) + 1$
36. $g(x) = -\left(\frac{1}{2}\right)^{x+2} + 1$
37. $g(x) = 2(x-2) + 1$
38. $g(x) = -3(x+4)^2 + 2$
39. $g(x) = \dfrac{3}{2x^2} - 1$
40. $g(x) = 3\sqrt[3]{4-x} - 1$
41. $g(x) = \dfrac{6x-7}{2x-3}$
42. $g(x) = 2\sqrt{-\frac{1}{2}x + 3}$
43. $g(x) = 2x^2 - 8x + 5$
44. $g(x) = 3^{\frac{1}{3}x} - 5$
45. $g(x) = \log_2(x-3) + 4$
46. $g(x) = -\left|4 - \frac{1}{3}x\right| + 2$